Baker Hughes

Sondex Wireline Limited

UltraWire Memory Control Unit

Control Unit for UMT Type 2 Tools

Operation & Maintenance Manual

Tool Code: UMU002
Document: MN-UMU002 EN
Revision: A
Date: 08 January 2020

Contents

1 Safety. 1-1

2 Overview. 2-1
2.1 Electronic Description 2-2

3 Specification. 3-1

4 Installation Procedure 4-1
4.1 Software 4-1

5 Operational Procedure 5-1

6 Maintenance 6-1

Appendix A Equipment & Spares..... A-1
A.1 Equipment Supplied..... A-1
A.2 Maintenance Equipment..... A-1
A.3 Items Not User Serviceable..... A-1

Appendix B Drawings & Parts Lists.....B-1
B.1 Mechanical Drawings B-1
B.2 Electrical Drawings B-1

References used in this Manual

Manual History

Contact Information

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1 SAFETY

Reference: Safety Information manual ***Safety Information***

In normal use, there are some specific instructions to safely handle this equipment. This information must be followed in addition to your specific Company, Local and International Rules & Regulations (Directives) in accordance to relevant national legislation.

Personnel must be qualified to operate or service this equipment. When the equipment is not installed, commissioned, used and maintained in accordance with the specifications provided, protection will likely be impaired.

This manual must be read and understood in conjunction with all other manuals that relate to the tools and equipment that comprise the overall system. Any potential hazards and required safety precautions that apply to any part of the overall system must be fully understood and adhered to before the equipment is powered up, used or serviced.

When this equipment is incorporated with other equipment into a system, the safety of that system is then the responsibility of the person who assembles the system.

Appropriate Personal Protection Equipment (PPE) must be worn at all times during the maintenance and service of this equipment. Protective glasses, shoes and gloves must be worn as a minimum along with recommended additional PPE referred to in the relevant tool manual.

Table 1-1 Warning Symbol Definitions

Symbol	Definition	Symbol	Definition
	General warning		**Explosive material**
	Radioactive material		**Laser source**
	Magnetic field		**Electrical hazard**
	Toxic material		**Hot surface**
	Flammable material		**Sharp element**
	Corrosive substance		**Finger/hand trap**
	Carcinogenic substance		**Trapped pressure**
	Heavy equipment		**Environmental hazard**
	Overhead load		**Falling objects**

Table 1-2 Prohibition Symbol Definitions

Symbol	Definition	Symbol	Definition
	General prohibition		**Do not touch**
	No access for people with active implanted cardiac devices		**Keep away from Magnetic Fields**

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Table 1-3 Mandatory Symbol Definitions

Symbol	Definition	Symbol	Definition
	General mandatory		Refer to instruction manual
	Connect an earth terminal to the ground		Disconnect mains plug from electrical outlet
	Wear eye protection		Wear safety footwear
	Wear protective gloves		Wear a face shield
	Wear head protection		Wear protective clothing
	Wash your hands		

Table 1-4 Other Symbol Definitions

Symbol	Definition	Symbol	Definition
	Electro-Static Discharge (ESD)		
	WEEE Disposal		Battery Disposal

Table 1-5 Hazard and Information Definitions

WARNING	**Potential risk of injury.**	
Caution	**Potential risk of minor injury or damage to equipment.**	
	DO NOT	Perform an action or process. This is an instruction that **prohibits** the action or process.
	DO	Perform an action or process. This is an instruction that **mandates** the action or process.

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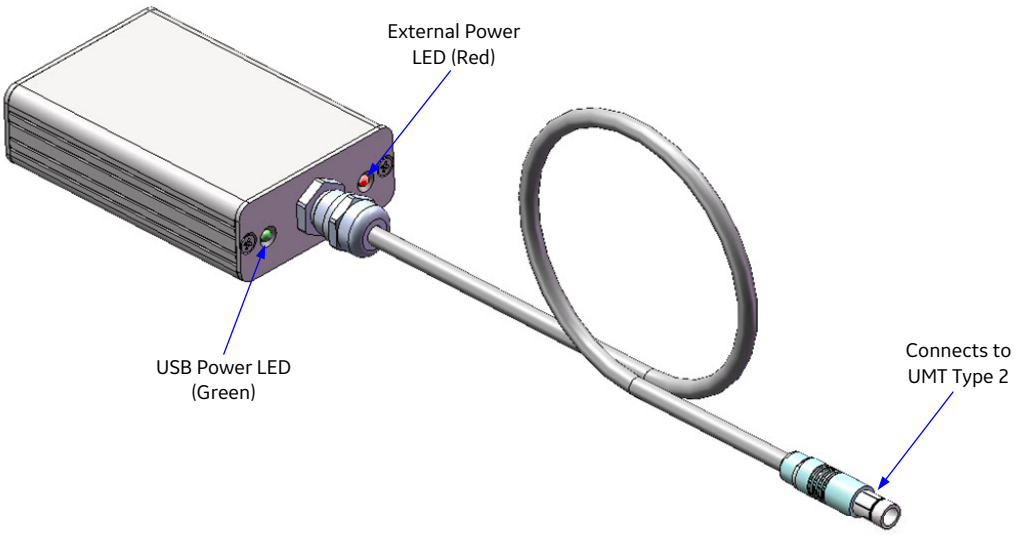
2 OVERVIEW

The UltraWire Memory Control Unit (UMU002) is used as an interface to connect Type 2 UltraWire Memory Tools (for example UMT007 & UMT008) to a PC (for example, Desktop, Laptop or Notebook) that has installed on it the UltraWire Memlog software. The UMU002 interfaces between the Low Voltage Differential Signalling (LVDS) connection inside the downhole tool and the USB connection that is part of a PC.

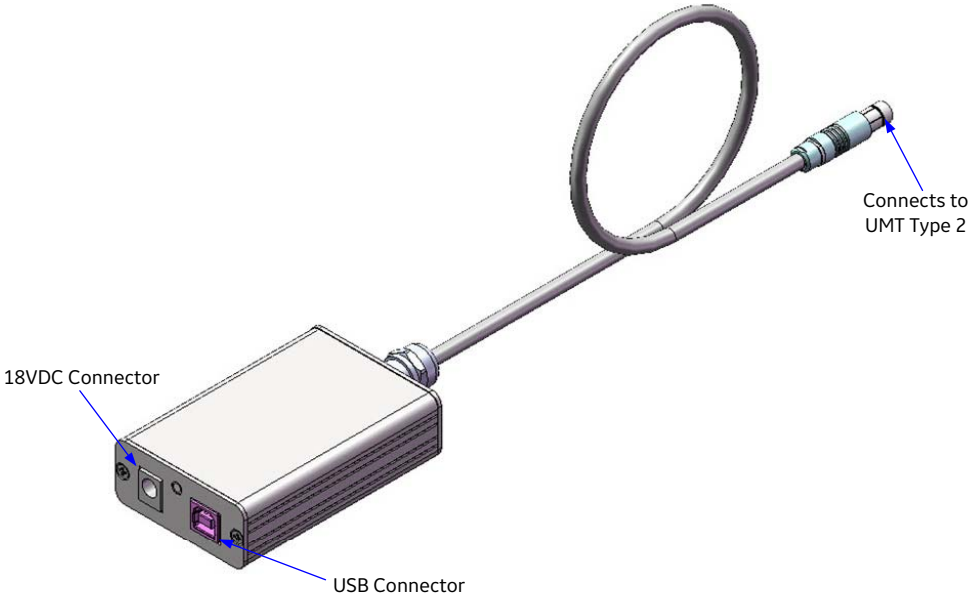


The UMU002 is used to either program a UMT Type 2 with a Logging Profile or transfer the Logging Data from the UMT memory to the PC when used with the USB Connection

In addition, the UMU002 allows complete toolstrings to be powered-up and checked-out on the surface prior to deployment downhole, but only when an 18VDC input is connected.

The UMU002 consists of a:

- **UMU Interface Assembly**



The UMU002 assembly houses the interface electronics. It has a USB 2.0 connector, a 18VDC power connector and an integral lead for connection to a UMT Type 2.
In addition, the UMU002 has a USB cable (P/N: W027-AB3M0) to link the USB 2.0 connector to the PC. When you only want to either programme or download the UMT Type 2, the USB connection from the PC will supply sufficient power to do this.

- **Mains Power Supply**
  When a complete toolstring is to be run from the PC for test purposes, the Power Supply

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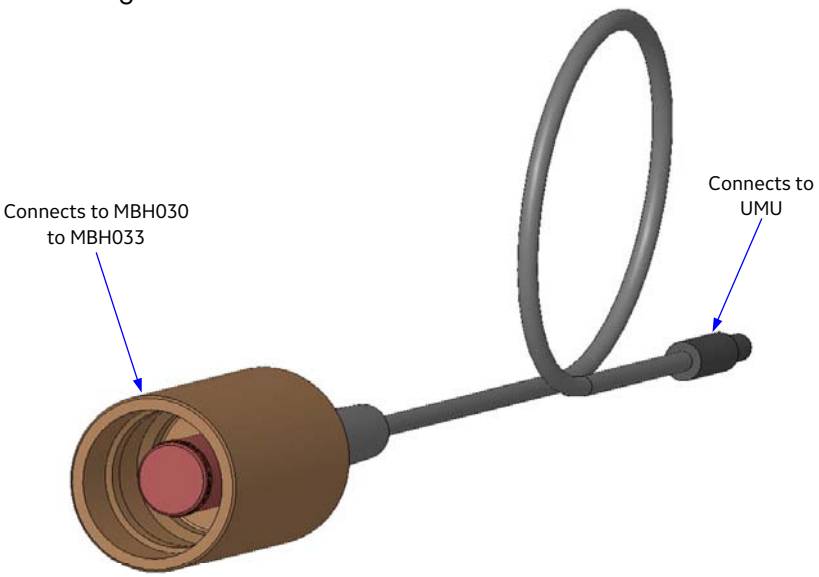
Unit (PSU) (P/N: 414342) can be plugged into the UMU002 and has enough power to run most toolstring combinations.



When the PSU is not available, there is a MBH/UMU Power Lead Assembly (P/N: 411024) that powers the UMU002 from a UMT Type 2 MBH Battery Unit through the same 18VDC power socket used by the PSU.

Note: It must be remembered that when a UMT Type 2 MBH Battery Unit is used, there will be a limit to the time this can be used dependant on the charge in the batteries.

The USB interface is 2.0/1.1 compatible, while the UMT007/UMT008 interface is full-duplex Low Voltage Differential Signalling (LVDS). The transfer of information is possible only when the UMT007/UMT008 is at the surface and is via the UMU002.

Although the UMU002 is USB 1.1 compatible, it will operate at a much reduced speed. Connection to a USB 2.0 port is strongly recommended.

The USB interface uses a standard Type 'B' receptacle that connects via a normal USB cable to the Type 'A' socket on a PC. The USB connector is marked on the box with a USB symbol (as shown on the right).

2.1 Electronic Description

The UltraWire Memory Control Unit (UMU002) provides:

- A 500kbaud asynchronous Low Voltage Differential Signalling (LVDS) command transmit channel to a UMT007/UMT008.
- 500kbaud asynchronous LVDS reply receive channel from a UMT Type 2.
- A 12Mbaud LVDS data download and monitor channel from a UMT Type 2.

The two receive channels use the same physical wires and are multiplexed as necessary.

Most of the functionality is provided by a USB driver chip that is programmed with the appropriate USB identifiers. Two asynchronous serial ports are connected to Low Voltage Differential Signalling (LVDS) level converters, the receive converter that is shared as above.

A MOSFET switch is used to disconnect the USB power from the UMT Type 2 when external power is connected.

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3 SPECIFICATION

Table 3-1 UMU002 Specifications

Parameter	Specification	Remarks
Explosive Operations	UMU002 **MUST NOT** be used.	Refer to ***Safety Information***, Section 3.3.
Unit Temperature Range:		
Transportation/Storage	-20°C (-4°F) to 85°C (185°F)	Also refer to PSU Module below
Operational	0°C (32°F) to 65°C (149°F)	Also refer to PSU Module below
Altitude Limitations	0ft (0m) to 6562ft (2000m)	
Operational Humidity	Up to 85% (no condensation)	
Environmental	For indoor use only	
Unit Dimensions:		
Height	1.0″ (25mm)	
Width	2.2″ (55mm)	
Length (cable not included)	3.4″ (86mm)	
Length (cable included)	26.8″ (680mm)	
Transportation Dimensions (fitted in Transit Case):		
Height	16.5″ (420mm)	
Width	7.1″ (180mm)	
Depth	13.8″ (350mm)	
Transportation Weight	9.3lbs (4.2kg)	Includes UMU, Cables & PSU module in Transit Case
USB:		
Version	USB 2.0	
Maximum Current	500mA	
Input:		
Maximum Voltage	20VDC	
Maximum Current	2A	
Output:		
Maximum Voltage	20VDC	
Maximum Current	2A	
EMC Classification	Class A Product	Refer to ***Safety Information***, Section 8.9.

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Table 3-1 UMU002 Specifications (Continued)

Parameter	Specification	Remarks
PSU Module:		
Temperature Range:		
Transportation/Storage	-20°C (-4°F) to 85°C (185°F)	
Operational	0°C (32°F) to 40°C (104°F)	
Dimensions:		
Height	1.3” (32mm)	
Length	4.5” (115mm)	
Depth	2.0” (50mm)	
Input:		
Voltage Range	100VAC to 240VAC	
Frequency	47Hz to 63Hz	
Maximum Current	1.7A	
Output:		
Voltage	18VDC	
Maximum Current	3.6A	

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4 INSTALLATION PROCEDURE

4.1 Software

Prior to connection of the UMU002, make sure the PC has the applicable UWMEMLOG software and the relevant USB driver software installed.

4.1.1 USB DRIVERS

The Driver Packages.zip is available to download from the ***Sondex*** software downloads page and can be found in the Drivers tab. Make sure the latest driver is installed. If not, uninstall and install the latest version. The required driver packages are:

- Windows Driver Package – Sondex FTDI CDM Driver Package (ftdibus) ([date] [version]).
- Windows Driver Package – Sondex FTDI CDM Driver Package (ftdiport) ([date] [version]).

To install the Sondex driver:

1. Select and run the Sxftdi.exe driver install program to pre-install the drivers on to the PC. Repeat for the driver Sxdxusb.exe.

2. Connect the UMU002 and apply power (for the first time). The PC will bring up the "Found New Hardware Wizard" notification.

3. The software will automatically be installed (select the appropriate driver if prompted).

4. To check the driver installation, open the PC Device Manager and view the Universal Serial Bus controllers. The UltraWire Memory Unit must be listed.

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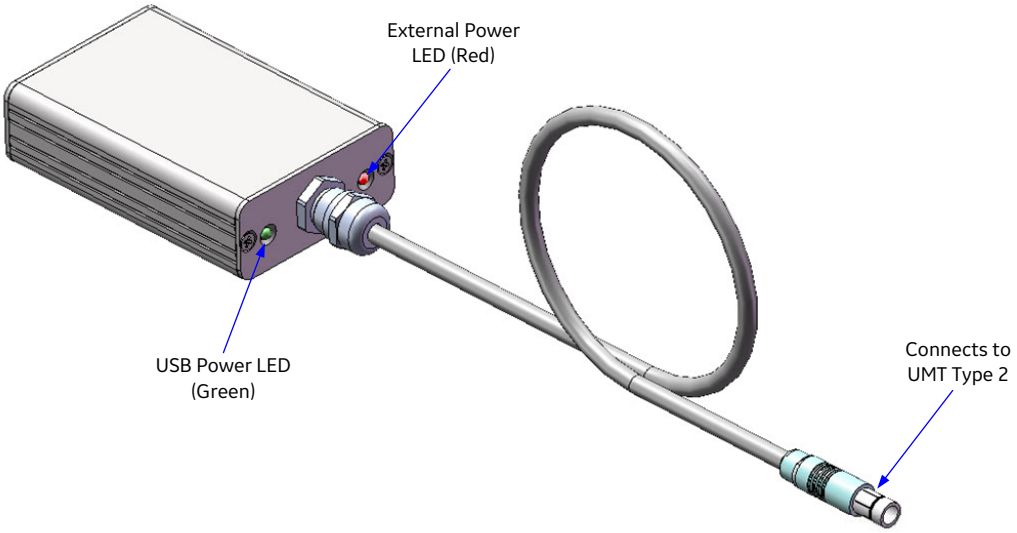
5 OPERATIONAL PROCEDURE

Communication between the UMU002 and the memory tool is via a Low Voltage Differential Signalling (LVDS) interface that uses a captive 600mm cable with a 7-Way connector that plugs into a UMT Type 2.

The USB power is sufficient **ONLY** to operate a UMT007/UMT008 for programming and data download. External power at a nominal 18VDC **MUST** be supplied when it is required to operate a toolstring. This uses a standard DC power connector marked '18V DC' on the UMU002 box. External power can be supplied by either the Power Supply Unit (PSU) supplied with the UMU002 or through the use of a suitable cable in conjunction with an MBH (for example the MBH/UMU Power Lead Assembly - supplied with UMU002). **ALWAYS** make sure the power is removed from the UMU002 and UMT007/8 tools during connection or disconnection.

When an external power unit is used, the UMU002 disconnects the UMT Type 2 from the USB power - the UMT Type 2 runs only on the 18VDC supply. However, the USB cable must remain connected to the PC.

Two LEDs are provided on the captive cable end of the UMU002 and operate as follows:



- A Green LED labelled 'USB Power' indicates that USB power is present.
- A Red LED labelled 'External Power' indicates that an external 18VDC power is present.

LED Indication	Power from...	Function
Green LED 'On'	USB power supply connected.	USB Interface chip and UMT use USB power. UMT **cannot** drive a toolstring.
Red LED 'On'	External 18VDC power supply connected - either a PSU (supplied with UMU002) or an MBH.	UMT powered from 18VDC. UMT **can** drive a toolstring.
Red & Green LEDs 'On'	Both USB and External power supplies connected.	USB interface chip powered from PC. UMT powered from the external power supply and can drive a toolstring.

Each tool in a toolstring requires a minimum level of input voltage (for example 13.5VDC or more) and when the external power supply to the UMU002 is less than this minimum level the individual tool can stop operation or deliver corrupt data. If the external 18VDC power is removed, all tools in the toolstring will cease to operate. However, with the USB cable connected between the UMU and the PC data transfer can still be made.

The USB power supply to the UMU002 is **ONLY** for programming and data transfer.

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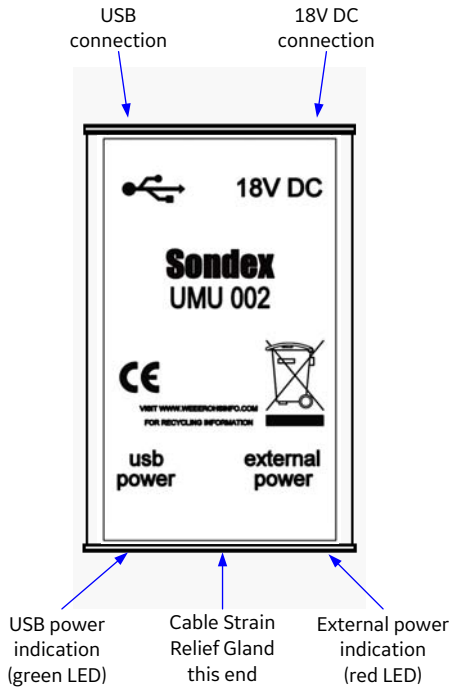
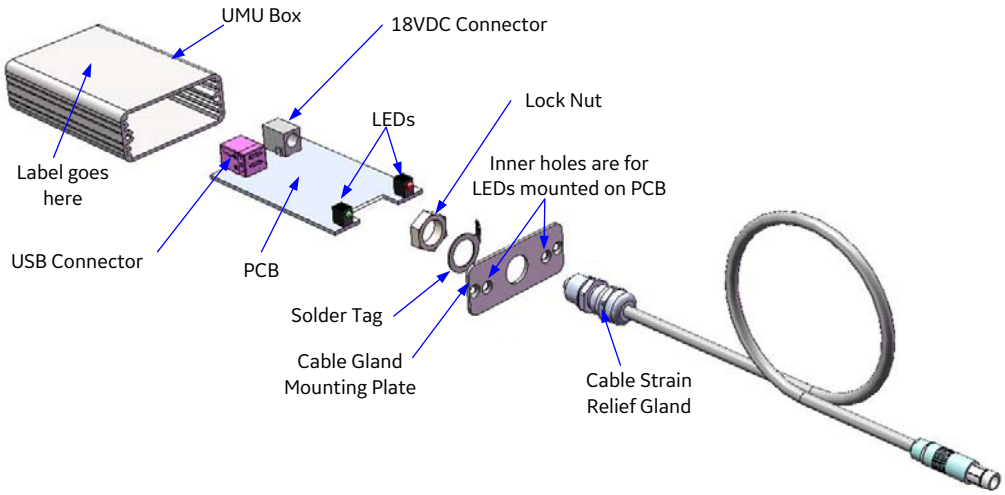
6 MAINTENANCE

Caution	**Electrostatic Discharge (ESD)** See ***Safety Information***, Section 8.6.

Reference: UMU002 General Assembly ***AD-421456***

The main cable assembly that is fixed to the UMU002, must **NOT** be taken apart for repair. Replace if damaged. **NO** repairs must be attempted/made to the UMU002 other than cable replacement.





Note: Make sure the Blue/Black and the Green/Black pairs are twisted together and are as short as is possible as these wires carry the high-speed data.

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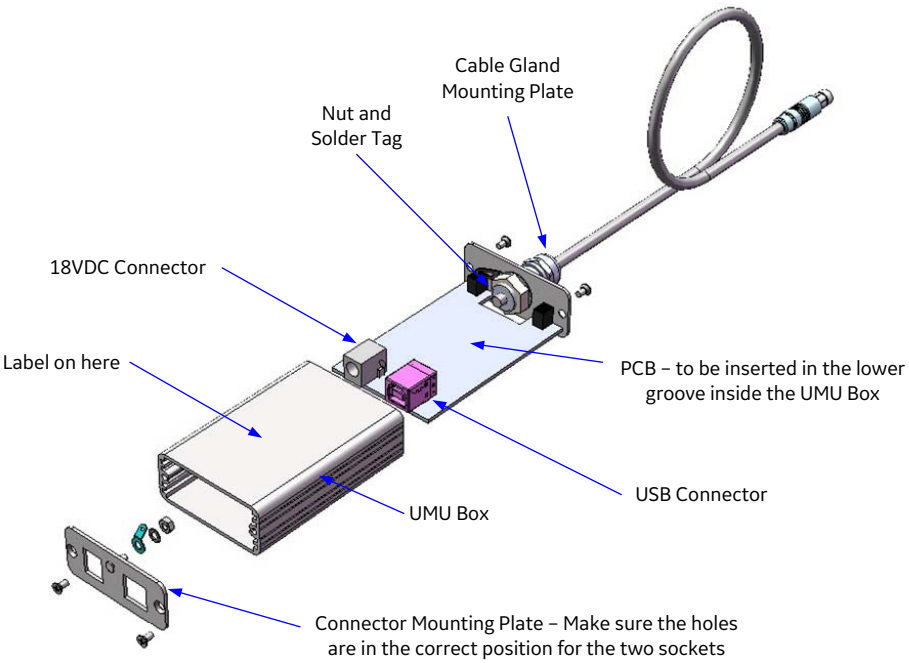

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APPENDIX A EQUIPMENT & SPARES

A.1 Equipment Supplied

Part No	Description	Qty	Remarks
UMU002	UltraWire Memory Control Unit, supplied with:	1	
411024	MBH/UMU Power Lead	1	
414342	PSU, I/P 90-264VAC, O/P 18VDC, 3.6A, 65W	1	
94112	Lead Mains (13A UK to IEC (3A Fused))	1	
94113	Lead Mains Supply (USA to IEC)	1	
94123	Lead Supply 2m Long 10A IEC To Schuko (EU)	1	
94130	Main Supply 2m Long (10A IEC P to IEC Skt)	1	
W027-AB3M0	Cable, USB, Type A to B, Length 3m	1	

A.2 Maintenance Equipment

A.2.1 CONSUMABLES

Part No	Description	Qty	Remarks
414575	SAC305/SAC387 Flux Cored Solder Wire	1	0.5kg reel

A.3 Items Not User Serviceable

Part No	Description	Qty	Remarks
421456	UltraWire Memory Control Unit Assembly	1	Except to replace the cable as detailed in ***Section 6***.

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APPENDIX B DRAWINGS & PARTS LISTS

B.1 Mechanical Drawings

Description	Drawing	Parts List
UMU002 General Assembly	***AD-421456-A***	See drawing

B.2 Electrical Drawings

Description	Type	Drawing
UMU002 Cable Assembly	Wiring Diagram	***WD-421456-A***

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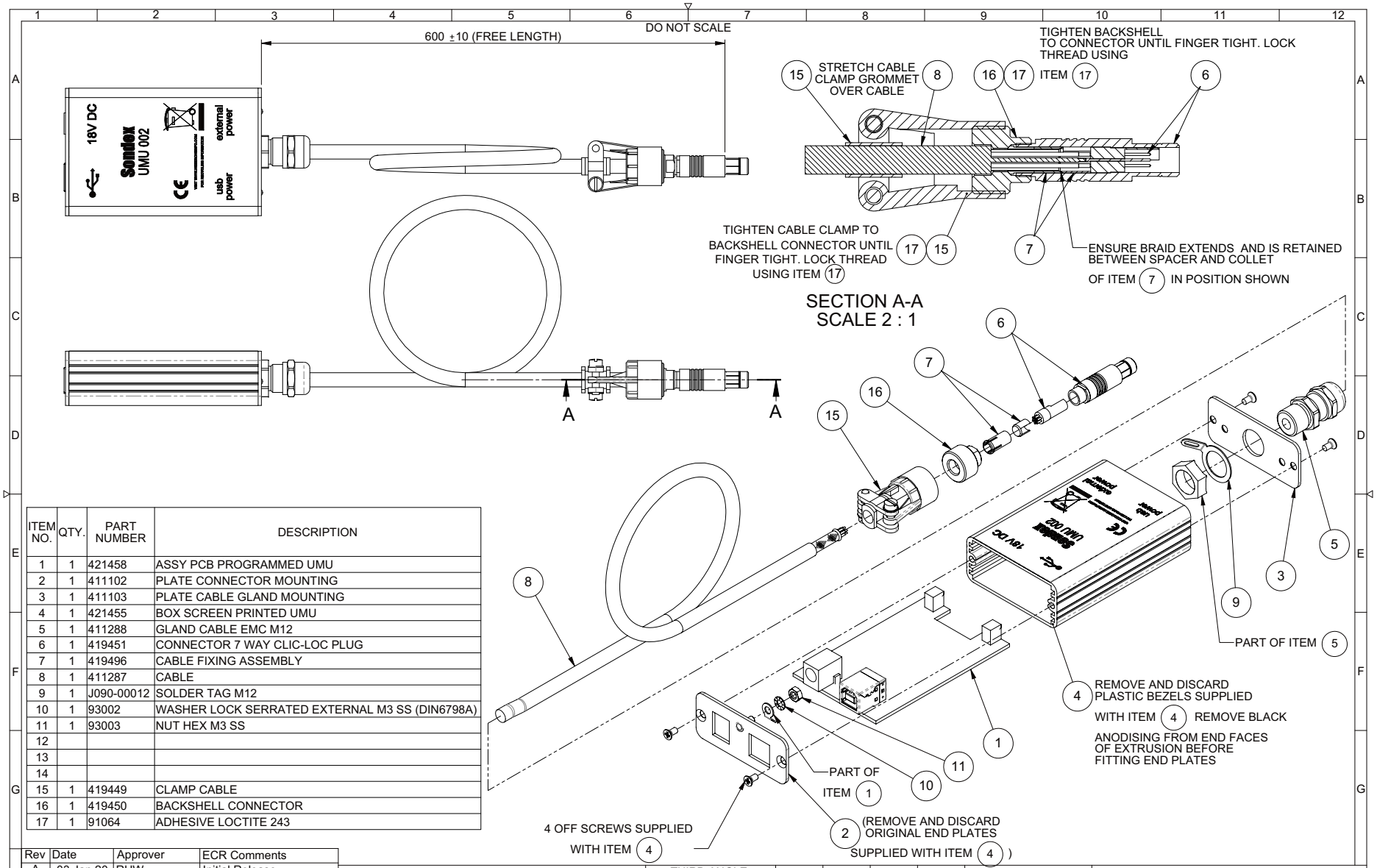

ITEM NO.	QTY.	PART NUMBER	DESCRIPTION
1	1	421458	ASSY PCB PROGRAMMED UMU
2	1	411102	PLATE CONNECTOR MOUNTING
3	1	411103	PLATE CABLE GLAND MOUNTING
4	1	421455	BOX SCREEN PRINTED UMU
5	1	411288	GLAND CABLE EMC M12
6	1	419451	CONNECTOR 7 WAY CLIC-LOC PLUG
7	1	419496	CABLE FIXING ASSEMBLY
8	1	411287	CABLE
9	1	J090-00012	SOLDER TAG M12
10	1	93002	WASHER LOCK SERRATED EXTERNAL M3 SS (DIN6798A)
11	1	93003	NUT HEX M3 SS
12			
13			
14			
15	1	419449	CLAMP CABLE
16	1	419450	BACKSHELL CONNECTOR
17	1	91064	ADHESIVE LOCTITE 243

Rev	Date	Approver	ECR Comments
A	03 Jan 20	RHW	Initial Release

UNLESS OTHERWISE STATED:
DRAWN TO BS 8888 & LOCAL STANDARDS PS166
GENERAL TOLERANCES IN ACCORDANCE WITH ISO 2768-mK
METRIC THREADS TO BE COARSE SERIES 6H/6g FIT
UNDERCUTS TO BS1936
SURFACE TEXTURE TO BE 1.6 µm MAXIMUM INTERNAL RADII 0.5

MARK THE COMPONENT IN / ON THE AREA IDENTIFIED THUS:

ALL MARKING SHOULD BE DONE PER PROCESS SPEC: PS-075

REMOVE ALL BURRS & SHARP EDGES.

THIRD ANGLE PROJECTION

ALL DIMENSIONS IN MM

USED ON: UMU002

MASS: 0.38 kg

MODEL'D	DRAWN	CHECK'D	APPV'D
GC	GC	ICJ	RHW
22 Oct 19	22 Oct 19	03 Jan 20	03 Jan 20

MATERIAL: SEE PARTS LIST

HEAT TREATMENT: NOT APPLICABLE

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TITLE: ASSEMBLY ULTRAWIRE MEMORY CONTROL UNIT

SCALE: 1:1

DRAWING No.: AD-421456

REVISION: A

SHEET 1 OF 1

A2

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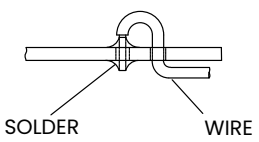
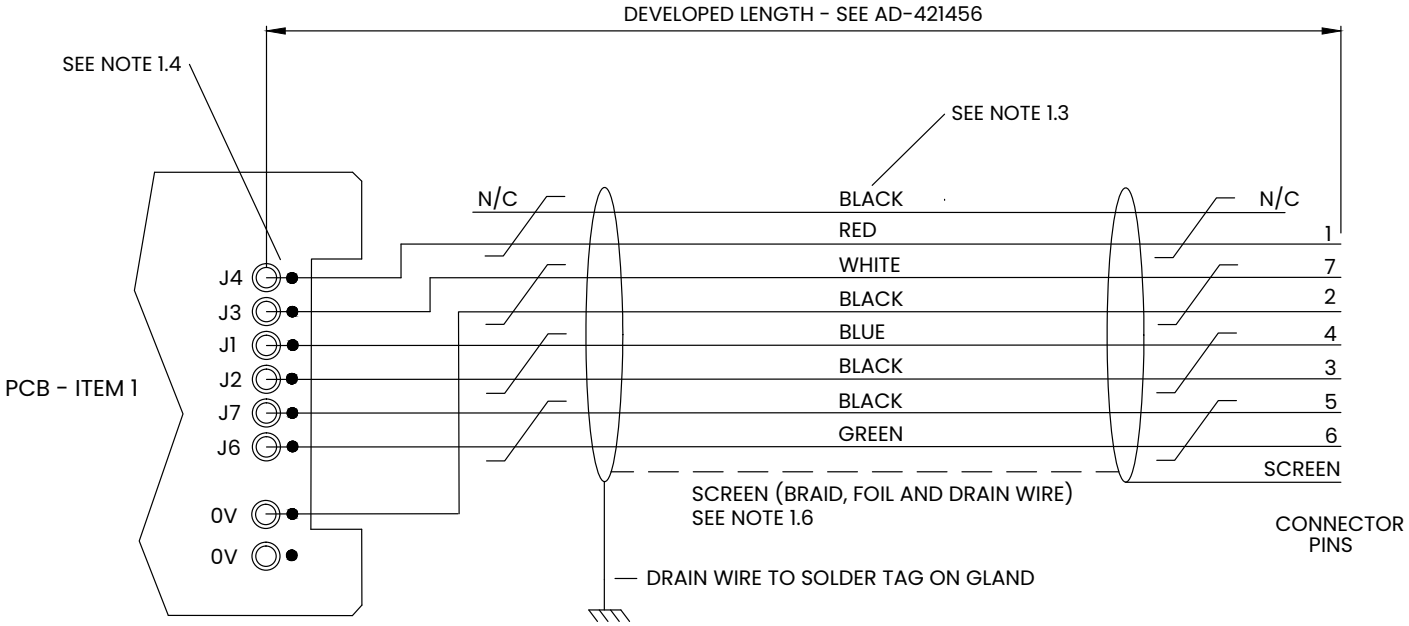
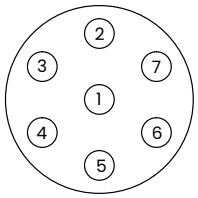
NOTES: -

1.1. FOR SOLDERING REFER TO PROCESS SPEC. PS117.
1.2. ACCEPTABILITY OF ASSEMBLY - SPEC IPC A 620 CLASS 3 & PS188.
1.3. THE BLACK WIRE TWISTED WITH THE RED WIRE, PART OF ITEM 8, IS NOT USED AND SHALL BE CUT BACK.
1.4. PASS WIRES THROUGH STRAIN RELIEF HOLES FROM SIDE 2 TO SIDE 1 OF BOARD THEN BACK THROUGH SOLDER PADS FROM SIDE 1 TO SIDE 2 OF BOARD ENSURING SOLDER COMPLETELY FILLS HOLE.
1.5. KEEP ALL TWISTED PAIRS AS SHORT AS POSSIBLE
1.6. ENSURE THAT SCREEN MAKES GOOD CONTACT WITH CABLE GLAND.





7 WAY CONNECTOR
ITEM 7



VIEW ON SOLDER BUCKET
SIDE OF CONNECTOR

NTS - ALL DIMENSIONS IN MM

REV	CHANGE	CHKD	DATE	APPD	DATE
A	ECR 182625 REFERS	DM	28-11-2019	CS	28-11-2019

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TITLE
UMU002
WIRING DIAGRAM

DRAWN	CHECKED	APPROVED
PC	DM	CS
DATE 28-11-2019	DATE 28-11-2019	DATE 28-11-2019

DRAWING No. WD-421456
REVISION A
SHEET 1 of 1

A3

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REFERENCES USED IN THIS MANUAL

- Safety Information manual (***Safety Information***).
- WEEE/RoHS Information: ***www.weeerohsinfo.com***.

MANUAL HISTORY

Date	Issue	Description	Auth	Chk	App
08/01/2020	A	Initial release.	SL	CS/RW	PR

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